Skill 2: Predict House Prices using Linear Regression

A real-estate company wants to estimate house prices from property characteristics. For a simple classroom example, we use the California Housing dataset available through scikit-learn.

In [2]:
```
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np
import pandas as pd
```

In [3]:
```
housing = fetch_california_housing(as_frame=True)
X = housing.data
y = housing.target
```

In [ ]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)
```

In [ ]:
```
lr = LinearRegression()
lr.fit(X_train, y_train)
```

Out[ ]:

▾ LinearRegression

▶ Parameters

In [7]:
```
y_pred = lr.predict(X_test)
```

In [8]:
```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("MAE :", round(mae, 4))
print("MSE :", round(mse, 4))
print("RMSE :", round(rmse, 4))
print("R2 :", round(r2, 4))
```

```
MAE : 0.5332
MSE : 0.5559
RMSE : 0.7456
R2 : 0.5758
```

In [10]:
```
coef_table = pd.DataFrame({
"Feature": X.columns,
"Coefficient": lr.coef_
}).sort_values("Coefficient", key=abs, ascending=False)

coef_table
```

Out[10]:

	Feature	Coefficient
3	AveBedrms	0.783145
0	MedInc	0.448675
7	Longitude	-0.433708
6	Latitude	-0.419792
2	AveRooms	-0.123323
1	HouseAge	0.009724
5	AveOccup	-0.003526
4	Population	-0.000002